

Factors that determine the reputation of private equity managers in developing markets

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Abstract

This paper analyzes the relationship between investors and private equity managers in order to identify the factors that affect the latter's reputation. Since there are no individual references about their past returns in developing private equity markets, the reputation of such players is thought to be linked to their capacity for obtaining new funds. Results provide evidence of the volume of investments recorded in the past, the ratio of portfolio companies to investment manager, the percentage of divestments carried out through initial public offerings and trade sales, the membership of the national private equity association and the size of funds under management as characteristics of the highest importance in raising funds.

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1. Executive summary

This paper analyzes the relationship between investors and private equity operators within the framework of the Agency and the Signaling Theories. It is designed to increase our understanding of the reputation-building mechanisms used by private equity operators to minimize the problems of information asymmetry that arise in their relationship with

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investors and that will allow them to attract funds from the latter. Reputation of managers is linked to track record, so in developed countries where this information is available, investors may decide to allocate their funds on that basis.

However, in countries where private equity markets are still in a process of attaining maturity, many management teams do not have any track record to show. The length of the investment–divestment cycle would limit the availability of these data due to the scarce number of fully liquidated funds. Even if data regarding returns were available, they would refer to a particular moment of time and not to time series because private-equity-backed firms are not quoted on official markets. In these cases, high-quality private equity operators are forced to send credible signals to the market in order to distinguish themselves from bad operators and raise additional funds.

This issue is of unquestionable importance in private equity. Venture capital and private equity firms are organized as limited partnerships, with the general partners having to raise funds periodically from limited partners in order to continue the activity. However, this question has not yet been addressed in the empirical literature. The aim of this paper is to provide evidence on the factors that characterize private equity operators' reputation in developing markets. In particular, the Spanish market is used as a framework for analysis. The unavailability of returns leads us to define a proxy variable for reputation. Due to the cyclical nature of private equity activity, most managers must seek funds regularly. Thus, we infer that the most reputed ones are those who raise most funds.

In this paper, two large groups of variables indicative of a reputation are identified: one related to the private equity activity, and a second based on external signs. Regarding the first group, results show that the investment activity is positively perceived by investors, so that managers who make more investments also raise more funds. However, investors penalize those managers who maintain a high ratio of portfolio companies per investment manager as expected future returns decline with low levels of management attention per investment. Results also show that managers who raise most funds are those who divest through initial public offerings and trade sales, which is consistent with the fact that these mechanisms are the most profitable ways of exiting. Regarding the second group of variables, membership of the national venture capital and private equity association is positively perceived by investors.

This paper also studies the different signaling patterns that could be used by private-sector and government-related operators. Results show that while the former are more concerned about building and maintaining a good reputation, the latter scarcely send signals to the market. The different behavior may be based on the diverse aims they were set up for. Managers for whom the majority of resources are provided by private-sector investors are more obliged to account for their performance to them. In the case of government-related operators, however, greater importance is given to economic and social objectives rather than to obtaining high financial returns.

In this way, the paper provides evidence that private equity managers send signals aimed at establishing their reputation, given their need to raise additional funds periodically. The results found in this paper may be useful both for private equity managers and investors. The former may now be aware of the variables that are considered by investors when committing funds to this activity. Also, the distinction between good and bad quality operators allows investors to allocate their money wisely.

2. Introduction

Venture capital is a type of business financing provided mainly through the acquisition of a stake in small and medium sized firms. This investment is temporary and typically represents a minority participation, since the principal aim of venture capitalists is the realization of capital gains upon divestment rather than corporate control. Initially, venture capital was intended as a way to finance firms in the early stages of development, since the lack of tangible assets or the high uncertainty inherent in the investment projects prevented those companies from obtaining long-term bank funding. Nowadays, the concept of private equity has become more extended and encompasses investment in any unlisted firm.

A characteristic element is that the funds invested by private equity operators (henceforth PEOs) are not their own but those of investors,² so PEOs act as intermediaries between suppliers and seekers of funds. This role proves important because PEOs are specialized financial intermediaries (Chan, 1983) able to reduce the level of information asymmetry between firms and their investors (Admati and Pfleiderer, 1994; Amit et al., 1998; Hellmann and Puri, 2002; Repullo and Suarez, 2004, among others). According to Gompers and Lerner (2001a) this can be achieved by the studies carried out by PEOs prior to the funding of firms, and their active role in the running of firms subsequent to such funding. This reduction in information asymmetry is the main reason why PEO intervention adds value (Black, 1998).

If we accept the necessary intermediary role of PEOs, the fundamental question then concerns the factors determining why investors commit their funds through certain PEOs rather than others. In spite of the natural importance of this issue, it has received little attention in literature. This paper aims to cover this gap. In a developed market, this question could be partially answered by the existence of a track record on previous funds. Investors would choose those PEOs who appeared able to generate higher return levels, given the risk involved. However, in markets where the private equity market is still developing, such information is not yet available and hence investors are expected to make their decisions on the basis of other criteria, such as PEOs' reputation (Schertler, 2002).

Thus, the principal aim of this paper is to determine the characteristics underlying the reputation of PEOs in developing markets, which will serve to explain their ability to attract new funds from investors. This relationship is analyzed from the theoretical support of the Agency and the Signaling Theories, given that their content perfectly fits the characteristics of the private equity market and the relationship established between PEOs and investors. This is tested in the Spanish market, which is still in the process of attaining maturity. However, the results obtained could be extended to other developing markets where data on returns are scarce or unavailable.

The paper is structured as follows. In the following section further details are given of the private equity activity and of the agents taking part. Two groups of signals derived from the premises of the Agency and Signaling Theories are also presented. Section 4 describes the sample, the methodology and the variables used in the empirical analysis. Some

² According to Sahlman (1990) the general partners typically provide about 1% of the capital. In the United States, pension funds were the main source. In Europe financial institutions are the main suppliers of funds for this activity (EVCA, 2004).

descriptive statistics are presented in Section 5. Section 6 presents the results obtained. The last section summarizes the main conclusions and discusses some implications for further research.

3. Theoretical framework

The Agency Theory is a suitable framework for analyzing the contractual relationships of private equity. According to Eisenhardt (1989) it seems reasonable to adopt this theory when investigating the problems within a principal–agent structure. In general, agency theory attempts to explain the relationship between one (principal) who delegates authority and decision-making power to another (agent), who performs that work (Jensen and Meckling, 1976; Eisenhardt, 1989). The private equity process allows the PEO to be viewed both as agent, in the relationship with the investor, and as principal, regarding the interaction with the portfolio company.

The bulk of the literature has studied the role of the PEO as principal and the entrepreneur as agent (Sahlman, 1990; Sapienza and Gupta, 1994; Smith, 1998; Reid, 1999; Kaplan and Stromberg, 2001, among others). The study of the role of the PEO as an agent of the investor has not received, by comparison, much attention. Only a few papers analyze several aspects of this relationship from the standpoint of the agency theory. Sahlman (1990) provides further insights on the contract between both parties and identifies the mechanisms used to overcome several conflicts stemming from this relationship. Osnabrugge (2000) compares the different approaches that venture capitalists and business angels may apply to deal with agency problems in their investee firms. According to this author, these differences have their origin in the greater pressure venture capitalists are under to behave competently for their fund providers. Osnabrugge and Robinson (2001) analyze the role that the fund provider may have in the structure, the investment behavior and informational strategies of the venture capital firm. Bankman and Cole (2001) study the underlying reasons for venture capital investments in a period where companies were overvalued and suggest that avoiding a loss of reputation on the PEO's side may have been one of them. Finally, Schmidt and Wahrenburg (2003) examine the effects of the reputation and bargaining power of PEOs on the design of partnership agreements. However, none of all these articles analyzes how PEOs establish their reputation in their relationship with investors, and this is the main contribution of this paper.

When the investor (principal) delegates decision-making power to the PEO (agent) in investing funds, it is understood that the agent will act in the best interest of the principal. Nevertheless, given the nature of this relationship, there is certain information that is only available to the PEO, thus giving rise to an information asymmetry problem that may exist in two forms. These are *hidden action*, that causes moral hazard, and *hidden information*, that gives rise to adverse selection (Eisenhardt, 1989; Amit et al., 1998). In these situations the agent does not act in the best interest of the principal, either deliberately, in the first case, or because of his inability to do so, in the second case.

With the aim of limiting the moral hazard problem, the relationship between investors and PEOs is formalized through a contract,³ containing various clauses aimed at offering

³ According to Sahlman (1990), these contracts are designed to protect investors from the likelihood of PEOs' making decisions against their interests.

incentives to PEOs to re-direct their behavior (Megginson, 2004). Specifically, Fama and Jensen (1983) point out that the agent and the principal sign a contract which specifies the agent's rights, the criteria for evaluating the results of his management, and his reward for good performance. Nevertheless, and given that the cost of drawing up such detailed contracts is very high, there are usually loopholes, giving PEOs leeway to act in their own interests. However, a bad fund management could damage the PEO's reputation and reduce its ability to raise new funds, thus limiting his opportunistic behavior.

The risk stemming from adverse selection is considerable in private equity, since PEOs have to carry out a wide range of duties and the final result depends upon the correct choice being made at all stages of the process. The costs of adverse selection can be reduced in two ways: improving the selection process of PEOs and identifying the signals sent by reputed PEOs. The former can be improved by increasing the information available in the market, either through an information search by investors or via publication of the PEO's track record. However, this is not available in economies where private equity is still developing. Therefore, the natural reaction of the market to the adverse selection problem in this context is signaling (Akerlof, 1970; Spence, 1973).

Thus signaling their reputation to investors helps PEOs reduce the problems of information asymmetry that arise in their agency relationship. The signaling literature assumes that there is information asymmetry between two parties. To overcome this information gap, the informed party uses tools to signal its value to the uninformed party. This allows the high quality informed party to be differentiated from the low quality informed one. Low quality types do not try to convince the market of its true value through signaling because of the cost involved. Only high quality individuals find it worthwhile to do so and thus appropriate the benefits of signaling.

The papers of Akerlof (1970) and Spence (1973) are considered the fundamental works on information asymmetry and signaling. Akerlof shows the problem known as 'market for lemons', whereby if cars of different quality levels are not distinguishable then they are all sold at a single price in the market. In absence of signaling this spurs the bad cars to flood the market and the good ones to get out of it. Spence describes signals as activities or attributes of individuals that alter the beliefs of, or convey information to, other individuals in the market. He shows, in the labor market context, how the workers with higher productive capabilities signal their quality to the employers by attending college.

Signaling theory can be extended to the information asymmetry problem that exists between PEOs and investors. PEOs have more information than investors about their true value. In particular, investors would like to know the ability of PEOs to generate substantial returns. As the track record is not known in developing markets, investors have to rely on the signals that PEOs send on their reputation. According to the signaling theory, the signaler is the PEO and the receiver is the investor. High quality PEOs must separate themselves from the low quality PEOs in order to raise money from investors and carry on their activity. The signals sent by PEOs may lead to a separating equilibrium among different types of PEOs. Thus it can be concluded that this relationship fits the premises of the signaling theory quite well.

The literature has analyzed a number of signaling mechanisms, such as the percentage of equity retained by the entrepreneur (Leland and Pyle, 1977), the level of debt (Ross, 1977), the level of capital expenditures (Trueman, 1986), the level of underpricing at initial public

offerings (Welch, 1989) or the choice of the underwriter (Carter and Manaster, 1990). In the context of the venture capital and private equity markets, a number of papers analyze within the signaling framework several aspects related to the information asymmetries that venture capitalists and investee firms face in their relationship (see Deeds et al., 1997; Davila et al., 2003; Elitzur and Gaviols, 2003; Janney and Folta, 2003; Busenitz et al., 2005, among others). However, to the best of our knowledge, no study has yet examined the relationship between PEOs and investors within this framework.

In order to reduce the problems of information asymmetry, the literature has considered an alternative mechanism to that of signaling: the screening device. This theory was pioneered by Stiglitz (1975). The idea is that the uninformed party can induce the other party to reveal his information by providing a menu of alternatives so that the choice depends on the private information of the other party. This would imply, for example, that investors would offer PEOs a contract with clauses that only high reputed PEOs will be willing to accept. The main difference between signaling and screening is that while in the former the informed party acts first, in the latter it is the uninformed party who does it. This paper focuses on the actions undertaken by PEOs to signal their reputation, so analyzing the screening devices is beyond the scope of this paper and constitutes an interesting area for future research.

Thus, by signaling their reputation PEOs contribute to reducing the problems of information asymmetry. In fact, as Schmidt and Wahrenburg (2003) point out, reputation can be seen as a security given by the agent to the principal. We agree with Rosenstein et al. (1990) that reputation can be linked to track record. In fact, Gompers and Lerner (1999) argue that the internal rate of return (IRR) of past venture capital funds is a nice measure of reputation. However, using this variable as a measure of reputation presents several methodological problems in this context. First, IRRs are usually held confidential by PEOs (Gompers and Lerner, 1999), thus posing problems to the access to this information. Second, and more importantly, this information is rather scarce in developing markets since PEOs have not had the time to close many funds. Third, even if this information was available, it would refer to a particular moment of time and not to time series because private-equity-backed firms are not quoted on official markets. This would not allow the exploiting of all the advantages of the panel data methodology, which is the one used in this paper. Moreover, and according to the signaling theory, since the value of signals erodes over time PEOs have an incentive to signal periodically so as to remain differentiated from low reputed PEOs. Thus an ideal measure of reputation should change over time in order to be credible and valuable.

An alternative proxy for reputation is, therefore, needed. As indicated by Smith (1999) and Schertler (2002), PEOs strive to establish a good reputation in order to attract investors to their funds. In fact, reputation determines the ability of managers to raise funds in developed private equity markets (Norton, 1995; Gompers and Lerner, 2001b; Janney and Folta, 2003; Schmidt and Wahrenburg, 2003). This should also be expected in developing countries. Thus the proxy for reputation will be the volume of funds raised. This proxy for reputation is in line with the one employed by Hsu (2004), who uses the number of funds raised as a measure of reputation. Also, this measurement for reputation is in the same spirit as the one employed in Sorescu and Subrahmanyam (2006), who rely on a measurement based on the amount of business generated by a bank in an initial public offering (IPO) to establish the reputation of IPO underwriters.

Thus the main aim in this paper is to find evidence of PEOs sending signals of their reputation that might be considered by investors. Two groups of signals that may explain the proxy for reputation are proposed in the following two subsections. Additionally, the last subsection analyzes the possibility of a different signaling behavior on the part of the private and government-related PEOs.

3.1. Signals related to private equity activity

3.1.1. Managers' experience

Managers' experience is one of the key variables that are assessed by investors when selecting a PEO in developed markets. It is expected that the greater the experience, the higher the reputation acquired and, therefore, the higher the amount of funds raised. Gompers and Lerner (1998) find that more experienced managers attract more funds. The liaison between experience, reputation and fundraising arises from the limited timeframe that PEOs have to invest, divest and give back the proceeds to the original investors, which is usually around ten years. Only those managers who succeed, earning a sizeable return, are able to raise more money easily from that moment onwards. The importance of experience is also highlighted by Kaplan and Schoar (2005), who find persistence in returns across a sequence of funds managed by the same venture capital firm. In this way, managers' experience can have an impact on the returns obtained and this may serve as a signal to investors of the reputation of managers. This proposition, however, may not hold in developing markets since the number of general partners with over 10 years of experience may be low and the number of new entrants may be high enough to offset the stake of the most experienced. Nevertheless, a positive relationship between the number of years of experience of the general managers and their reputation remains to be tested, even in developing countries.

3.1.2. Investment activity

Provided that in developing markets there is a limited number of managers with over 10 years of experience, most managers may signal their reputation through the different tasks they perform in their activity. Regarding investment activity, this is a complex process that takes a long time to complete, and there is fierce competition among PEOs. Therefore, those managers who finally succeed in negotiating and adding new investments to the portfolio will be considered by investors as highly skilful. In this line, Bankman and Cole (2001) find that even when PEOs know that investments are overvalued, they close deals so as not to lose reputation. Moreover, with each additional investment, the PEO extends his information network, hones his ability to appraise potential investments and gains experience in structuring deals or monitoring entrepreneurs (Sorenson and Stuart, 2001). Thus, it is more likely that PEOs with investment experience will acquire the skills needed to help investee firms to be successful, which is a powerful contributor to PEO reputation (Hsu, 2004). Therefore, the volume of investments made by PEOs may be a signal highly regarded by investors, who may thus distinguish between high and low reputed PEOs.

A second signal related to the investment activity refers to the fact that there may be an optimum number of firms in a PEO's portfolio (Bernile et al., 2003; Kanninen and Keuschnigg, 2003a,b; Cumming, 2006). In this sense, Keuschnigg (2004) states that PEOs invest in a limited number of firms due to the time required in the following up of each

investment. The managerial assistance provided by PEOs has proved to be important since it increases the probability of success of investee firms (Norton, 1995; Sapienza et al., 1996; Schmidt, 2002; Kanninen and Keuschnigg, 2003a,b; Walz and Cumming, 2004). Therefore, a decrease in management assistance could negatively affect the returns obtained by PEOs and a negative relationship between the number of investments made and the PEO's reputation is expected.

However, it is possible to increase the number of investments in the portfolio without reducing managerial assistance just by increasing the number of investment managers. Therefore, the variable that should be considered is the number of portfolio companies for each investment manager. In this line, Manigart et al. (2002) find that the required return of PEOs is higher when fewer investments per manager are carried out. Walz and Cumming (2004) find that a large number of portfolio companies per investment manager reduces future returns obtained by PEOs. Thus in order to signal their reputation, PEOs maintain a low ratio of portfolio companies to investment manager, and a negative relationship between these two variables is expected.

3.1.3. Divestment activity

Investors may also consider the ability of PEOs to divest. In this sense, Gompers (1996) shows that younger PEOs tend to launch IPOs at an earlier stage than more established ones, in an attempt to build a good reputation. Therefore, divestment opportunities have an impact on the operators' reputation. By divesting, PEOs can materialize the capital gains obtained and reward investors for the funds provided. However, the mechanism used to divest is also of importance. According to Ali-Yrkkö et al. (2003) PEOs build a reputation by the returns obtained in divestments, which in turn influences their ability to raise funds. IPOs and sales to an industrial or financial investor are assumed to provide the best returns.⁴ In fact, the bulk of capital gains earned by a fund is usually obtained from a small subset of the portfolio companies, namely those that are exited via an IPO or sale to another company (Hochberg et al., in press). However, excluding the United States (US), IPOs are rather scarce, with the latter being the main exit way in most countries. This is a credible signal that allows investors to identify high skilful PEOs, since low skilled ones may find it rather difficult to divest through these mechanisms. Therefore, a positive relationship between the proportion of investments that are successfully exited by means of IPOs or trade sales and the PEO's reputation is expected.

3.2. Signals related to external signs of reputation

Besides the signals related to the private equity activity, there may actually be other external signs used by PEOs to establish their reputation. These are as follows.

3.2.1. Membership of a national sector-related association

It may be considered as a signal whether the PEO belongs to a national sector-related association. The reason is that managers who are full members of the national venture capital and private equity association have to carry out their activity under the accomplishment of a

⁴ For an extensive review of venture capital exits see Cumming and MacIntosh (2003).

series of rules which are set up under a code of conduct. Furthermore, PEOs become more visible once they belong to the national association, thus diminishing the problems of information asymmetry. Therefore, a positive relationship between membership of the national venture capital and private equity association and the PEO's reputation is expected.

3.2.2. *Specific legislation*

Another reference relates to the way in which rules affecting the industry could exert an impact on the reputation of PEOs. In most countries there is specific legislation that provides fiscal incentives to investors committing funds to this activity. In order to be eligible (it is not mandatory), PEOs activity must be subject to monitoring by a state agency. It is considered that the increased visibility resulting from that monitoring, plus the commitment to carry out the activity under the aegis of the specific regulation currently in force, may signal the PEO's honesty and competence. Therefore a positive relationship between enrolment in the government's private equity programme and the PEO's reputation is hypothesized.

3.2.3. *Geographic location*

The geographic location of the PEO in important economic areas may also be relevant. This would be particularly important for PEOs aiming to attract non-resident investors for investment opportunities in the country. In the US, for instance, the biggest share money is raised by PEOs located around Silicon Valley and Boston's Route 128. The most highly reputed PEOs are located in those areas. Regarding the European experience the 2004 Directory of the European Private Equity and Venture capital Association, EVCA, shows that the highest number of PEOs choose the capital city as location for their headquarters (EVCA, 2004), leading us to expect a positive relationship between the location of the headquarters in the capital city and the PEO's reputation.

3.3. *Control variables*

Additionally, some control variables, which can explain the volume of funds raised regardless of the reputation of the operator involved, are also considered. These variables refer to the amount of capital under management by the PEO, the amount of funds still available for investment, economic growth, capital gains taxation, interest rates and returns obtained in the stock market.

The capacity to attract new funds may also depend on the amount of capital under management, that is, larger venture organizations would raise larger funds. Gompers and Lerner (1998) find evidence in this sense. These authors also argue that size might also measure the need to raise larger funds, because in order to keep all of the employees working at full capacity, the minimum fund size needed is substantially greater in these cases. The amount of capital awaiting investment is also considered important due to the cyclical nature of the private equity process. When this figure is high, investors are expected to provide fewer additional resources since this would increase the amount of funds awaiting investment.

Similarly, it should be noted that the volume of funds raised is potentially affected by other macroeconomic factors. The economic growth variable registers growth in gross domestic product (GDP) and it has also been considered in other works (Gompers and Lerner, 1998; Jeng and Wells, 2000). According to Gompers and Lerner (1998), if the economy is growing

quickly, the volume of funds raised may increase. In the same vein, Gompers and Lerner (1998) include capital gains taxation, even though Poterba (1989) finds evidence that this variable exerts an impact on the demand for venture capital rather than on supply.

Lastly, interest rates and stock market returns could also affect the volume of funds raised. Bonds and quoted shares are alternative investments to private equity, so if interest rates or stock market returns rise, the attractiveness of investing in this activity may deteriorate. Thus, a negative impact on fundraising is expected in both cases. Interest rates are considered as in Gompers and Lerner (1998). Nevertheless, we consider ten-year bonds in substitution of treasury bills because the term is more comparable to the length of the private equity cycle. Regarding stock market returns, Gompers and Lerner (1998) expect, however, a positive relationship on fundraising due to the significant importance of venture capital exits through IPOs in the US. On the contrary, the limited importance of exits through the stock market in Europe may lead to a negative impact on fundraising, since quoted equity is an alternative asset class that competes with private equity.

3.4. The structure of the market

The birth of venture capital and private equity markets in some countries was supported by government agencies (Manigart et al., 2002; Martí, 2002; Leleux and Surlemont, 2003). Some of the early PEOs were even set up with backing from multilateral institutions, i.e., the World Bank. This support may have some influence on the variables considered as possible signals, since these may not hold for government-related funds. In this type of funds raising new commitments will be determined by the priorities of the government-related institutions promoting them. Thus, in this case, establishing a reputation is not considered a priority because raising funds in the market is not foreseen. Indeed, the existence of this type of firm can only be justified by the lack of interest shown by private operators in certain regions or sectors. As Manigart et al. (2002) point out, government-supported funds are even willing to trade return for industry-employment and regional-development goals.

In our view, this could be the case for one-period players, but not for multi-period operators since the cyclical nature of private equity investment requires new inflows of funds periodically. Thus there might be some sort of signaling to justify an increase in contributions from government-related promoters. Therefore, even government agencies are interested in committing additional funds only if they positively assess the work done by the PEOs that managed previous funds. All PEOs, including those backed by government agencies, should therefore signal their reputation to their fund providers. However, it is interesting to test whether the types of signals sent differ between private-sector and government-related funds.

4. Research method

4.1. Sample

The sample covers the population of PEOs in the Spanish market during the period 1991–2003, in which 118 PEOs were operating. These data were obtained from the database of J. Martí (*U. Complutense*), who is the official source of information of the Spanish Private Equity Association (named ASCRI) and the European Private Equity and Venture Capital

Association, EVCA, regarding the Spanish data. Several PEOs, however, were excluded from the sample: i) 2 small PEOs who did not provide complete information on the variables related to the private equity activity; ii) 9 PEOs who handle pan-European funds, as the factors that drive reputation could differ depending on the country where funds are raised; iii) 6 PEOs born in the year 2003 since the empirical models proposed include lagged variables. The final sample refers to 101 PEOs, which represents 85.59% of the population and provides a total of 753 observations. The sources of information for the variables not related to the private equity activity are as follows: i) GDP (National Statistical Institute of Spain); ii) capital gains tax rate (Domestic Tax Laws); iii) interest rates (OECD Economic Outlook); iv) stock market returns (EcoWin).

4.2. Analysis

The panel data methodology is used in the empirical analysis, as data refers to time series observations for a sample of individuals. This methodology presents several advantages (see [Arellano and Bover \(1990\)](#), [Baltagi \(2001\)](#) for a review), of which the main one is that it enables us to control unobservable individual heterogeneity. Specifically, and in order to test the signals proposed, the empirical model is as follows:

$$\begin{aligned} &\text{Reputation}(\text{proxied by fundraising}) \\ &= f(\text{Previous private equity activity, External signs, Control}) \end{aligned}$$

The annual volume of funds raised, which is the dependent variable, takes positive values when PEOs raise funds and zero otherwise. Moreover, a high proportion of observations equal zero since fundraising is a time-consuming task and it takes 3 to 4 years to fully invest the amounts committed by investors. In this context, ordinary least-squares do not provide consistent estimations and more complex methodologies have to be applied. There are two approaches that deal with this problem.

The first methodology considers that data are censored, so that only observations beyond a threshold point are observed, zero in this case. The Tobit model ([Tobin, 1958](#)), and in particular the Tobit model type-I ([Amemiya, 1985](#)), should be used in this context. The second methodology assumes that zero value observations could instead be driven by the particular data generating process, implying that the Tobit methodology is not appropriate ([Sigelman and Zeng, 1999](#)). The latter approach seems more suitable here, because the zero values are driven either by the PEO's decision not to raise a new fund in a given year or by the PEO's inability to succeed in a fundraising process. Thus the decisions of individuals that cause the zeros should be modeled under this approach ([Maddala, 1992](#)). In this context, a standard regression will not provide consistent estimations because of the existence of a sample selection bias and a model of sample selection, which is known as the Tobit model type-II ([Amemiya, 1985](#)), should be applied.

The latter model allows different data generating processes for both the zero and the positive observations. In this model, two equations are estimated: a selection equation and a main equation. The [Heckman \(1979\)](#) methodology is usually employed to estimate this model, which gives consistent estimates of the parameters in the main equation via a two-step technique. In the first step, the selection equation is estimated and a new term, which is the conditional expectation of the error term, is generated. In the second step, this term is added as a new

regressor into the main equation with the aim of achieving a zero conditional expectation, and the main equation is estimated. The Heckman methodology has been extended to the panel data case (Wooldridge, 1995; Kyriazidou, 1997; Vella and Verbeek, 1999, among others). In this paper, an extension of the Heckman methodology based on the techniques developed by Verbeek and Nijman (1996) and Vella and Verbeek (1999) is used. The idea consists of eliminating the selection bias by incorporating two additional regressors into the main equation, which will be referred to as $s1$ and $s2$, hereafter. The Heckman methodology is more appropriate than the Tobit technique if one or both of these two elements become significant.

Thus the two equations in this context are as follows. The selection equation, which is a random effects probit model, measures the probability that a given PEO may raise a new fund in a given year. The endogenous variable ($Pnfund_{it}$) equals one in that case and zero otherwise. The structure of this equation is as follows:

$$Pnfund_{it} = Vpe_{it-1}'\beta + Vext_{it}'\gamma + Vcontrol_{t-1}'\lambda + v_i + v_{it}$$

where Vpe_{it-1} seeks to explain the reputation of PEOs from the four references related to the private equity activity commented on before; $Vext_{it}$ attempts to measure the PEO's reputation by means of the three external signs; and $Vcontrol_{t-1}$ refers to the control variables.

The main equation, which is a linear random effects model, measures the amount of new funds raised by a given PEO each year ($Nfund_{it}$) and takes the following form:

$$Nfund_{it} = Vpe_{it-1}'\beta + Vext_{it}'\gamma + Vcontrol_{t-1}'\lambda + \omega_1 s1_i + \omega_2 s2_{it} + \eta_i + \varepsilon_{it}$$

In both equations, all variables in the first group and available capital are lagged one period. This is justified because most of the information regarding those variables is available with a lag. The brochure would be based on last year's investment and divestment activity. Similarly, it would take some time to investigate the PEO's previous experience. On the other hand, fundraising is a long lasting time consuming task, with one year being the average period of time required to raise a new fund. Furthermore, investors themselves devote some time to assessing the information received from PEOs and thus decide their participation in a given fund. Whatever the cause, this lag could explain the vertical shape of the short term supply of funds, as formulated by Gompers and Lerner (2003).

As a final remark, it is worth noting that severe problems of multicollinearity may arise if the same variables are included in both equations. In order to avoid this problem, the selection equation should contain at least one variable that is not included in the main equation, which is known as an exclusion restriction. However, if a variable in the main equation is wrongly excluded an artificial sample selection bias will be detected. The reason is that the excluded variable would affect the independent variable in the main equation through the terms $s1$ and $s2$, so caution should be taken.

4.3. Variables

4.3.1. Dependent variable

The dependent variable is the annual volume of funds raised by each PEO. This variable refers to the total amount committed to a fund, although this will not all be disbursed on day one by investors but only as money is called for further investments. Thus, it includes the

total amount that PEOs receive for investment over the life of the fund. For example, if a given fund raises €40 million in 2000, this will be the quantity considered as the volume of funds raised in 2000, although it would be invested and disbursed in several years from the year 2000 onwards. This variable is expressed in constant 1991 € thousand. Some descriptive statistics of this variable are shown in Section 5.

4.3.2. Independent variables

The set of independent variables has been divided into two groups. The first one refers to the signals related to the private equity activity. This group includes managers' experience and managers' investment and divestment activity. The managers' experience records the number of years' experience of the team directing the private equity firm. This variable is considered, rather than the number of years that the institution itself has been in existence, because many of the recently created PEOs have built up their teams from experienced directors coming from other older firms. This measure of experience based on the firm-level rather than the fund-level assumes that the experience acquired in the running of one fund carry over to the firm's next fund. Regarding the investment activity, the first variable included is the amount invested in the previous year by a given PEO. The second variable registers the ratio, in the previous period, between the number of portfolio companies and the number of investment managers. Regarding divestment activity, data on the amounts divested the previous year by the PEOs are considered. It was possible to identify the volume divested through IPO and sale of the stake to third parties. These data, however, refer to values at cost instead of the real value of the transaction, as the latter was impossible to obtain. Nevertheless, assuming that divestitures through IPO and trade sale provide the best returns, a ratio of the volume of divestments through IPOs plus trade sales to the total amount divested is introduced as a signal of successful exiting. All numerical variables referring to investments and divestments are expressed in 1991 € thousand.

The second group of variables includes the three external signs. The variables in this group refer to factors such as: i) belonging to acknowledged private equity associations; ii) activity regulation; and iii) geographic location. All these effects are approached through dummy variables that equal one if the PEO: i) belongs to a sector-related association; ii) has signed up to the Law of Private Equity currently in force; iii) is based in Madrid. Regarding the first variable and in the particular case of Spain, the national association is named *Asociación Española de Entidades de Capital Riesgo* (ASCRI). ASCRI's assembly accepts or rejects candidates to become full members on the basis of their commitment to comply with sets of rules on the carrying out of their activity. The second variable refers to the industry regulation. The first legislation regulating private equity in Spain dates back to 1986. It established an authorization process for PEOs wishing to carry on their activity under the aegis of the law. Registered PEOs could take advantage of several fiscal incentives. The enactment of the Law 1/1999 considerably improved the legal and fiscal treatment of private equity. Under the new legislation, the legal powers for authorizing, registering and controlling are entrusted in the Spanish Securities Exchange Commission (*Comisión Nacional del Mercado de Valores*, CNMV), which is the agency in charge of supervising the Spanish Stock Markets and the activities of all the participants in those markets. This agency at present monitors the activity of registered operators. The last variable considers the geographic location of the PEO's headquarters in Madrid, which is

Spain's capital city. The large number of investors and, particularly, the amount of resources being managed in the country's capital may be the result of the higher reputation accruing from being located in the capital city. Descriptive statistics of all independent variables are offered in Section 5.

4.3.3. Control variables

The control variables are defined as follows. The size of the institution is included via a dummy variable that distinguishes between large or small size. Those PEOs who manage more than €60 million, at any given time, are classified as large size investors. The amount of capital awaiting investment represents the volume of funds still available for investment by the PEO in the previous year and is expressed in constant 1991 € thousand. The economic growth registers real GDP growth in Spain in the previous year. The tax rate considered is the top marginal capital gains tax rate each year. The interest rate variable refers to the ten-year bond return in the previous year and it is intended to match the average holding period in private equity investments. Finally, the stock market return refers to the previous year's annual return on the Spanish value-weighted market index (IBEX-35).

5. Descriptive statistics

Table 1 highlights the number of PEOs operating in Spain and its increase on a yearly basis. This number differs over the sample period, with a maximum number of 81 operators in 2002. It should be noted that the difference in the number of PEOs for two consecutive years does not always represent the increase in this figure, as there could also be PEOs that leave the market. This figure rose sharply for the years 2000 to 2002, where a high number

Table 1
Number of private equity operators in Spain

Year	Whole sample		Nature of funds					
	Number of PEOs	Increase of PEOs	Private-sector			Government-related		
			Number	Frequency (%)	Increase	Number	Frequency (%)	Increase
1991	44	8	25	56.82	7	19	43.18	1
1992	45	1	27	60.00	1	18	40.00	0
1993	53	8	32	60.38	4	21	39.62	4
1994	53	2	32	60.38	2	21	39.62	0
1995	53	0	32	60.38	0	21	39.62	0
1996	51	0	30	58.82	0	21	41.18	0
1997	49	1	29	59.18	1	20	40.82	0
1998	54	6	35	64.81	6	19	35.19	0
1999	54	5	36	66.67	5	18	33.33	0
2000	64	13	45	70.31	13	19	29.69	0
2001	74	12	55	74.32	11	19	25.68	1
2002	81	9	61	75.31	8	20	24.69	1
2003	78	6	58	74.36	5	20	25.64	1
Total				64.75			35.25	
Stability				97.26			90.14	

of PEOs were created in the new economy era. It is interesting to note that the new ones created were, basically, private-sector-related PEOs.

This table also shows the distribution of PEOs on the basis of the private or government-owned nature of their funds. There are 64.75% of observations belonging to the former and 35.25% to the latter. After some initial activity in the private equity sector in which the government played the principal role in Spain (Martí, 2002), the decreasing weight of government-related funding throughout the period considered can be noted. This fact characterizes the development of venture capital and private equity markets and it can be observed in several other European countries. The ‘Stability’ figure refers to the permanency in a given category in the period considered. That is, the 97.26% figure means that conditional on a PEO being private for a given year it remains being private in 97.26% of its observations. This figure is 90.14% for government-related PEOs. Thus, the stability in both categories is very high, with the corresponding effect on the validity of the results of the empirical analysis when a distinction is made between those groups.

Table 2 describes how PEOs are classified regarding the three external signs proposed, the PEO’s experience and the size of the organization. Taking the whole sample, it can be observed that 68.39% of the observations belong to operators that are members of the national private equity association (ASCRI). However, less than half of the organizations have signed up to the Law of private equity in force and have their headquarters located in Madrid. As for experience, 67.19% of observations belong to operators who have up to 10 years of experience in the industry, 8.44 being the average number of years of experience, with a median of 8. Finally, the percentage of large PEOs represents 19.52% of observations. This figure, however, has been increasing over time, from a percentage of 6.82% in 1991 to a value of 32.05% in 2003, which was to be expected in maturing markets, such as the Spanish one.

Considering the different types of PEOs according to the nature of funds, the percentage of PEOs who are members of the national private equity association is more or less the same between the private and the government-related PEOs. However, there are substantial differences between them in the remaining characteristics. The percentage of private-sector PEOs who have signed up to the Law of private equity in force and whose headquarters is in Madrid is much higher. Regarding size and experience, private-sector PEOs are of a larger

Table 2
Distribution of PEOs according to several characteristics

Characteristics of PEOs		Frequency		
		Whole sample (%)	Private-sector (%)	Government-related (%)
National private equity association	Ascription	68.39	65.19	74.61
Private equity law	Enrolment	46.75	55.33	30.08
Location of headquarters	Capital city	41.83	55.33	15.63
Managers’ experience	0–5 years	37.18	47.89	16.41
	6–10 years	30.01	30.18	29.69
	11–20 years	28.43	20.32	44.14
	>20 years	4.38	1.61	9.76
Size of private equity organization	Large	19.52	22.13	14.45

size, although the government-related PEOs are the ones with the most experienced managers, as this activity was first carried out by government-related PEOs. While the mean number of years of experience is 6.63 for the private-sector PEOs, this figure is 11.95 in the case of the government-related ones. The median figures are 6 and 11, respectively.

Table 3 shows some descriptive statistics of the independent numerical variables considered. On average, PEOs invested nearly € 5 million per year, in 1991 €, this figure being € 5.8 million for the private-sector-related PEOs and € 3.3 for the government-related ones. The average number of portfolio companies per investment manager is 2.98. Nevertheless, this figure is higher for government-related ones, which is in line with the discussion in Section 3. Government-related PEOs are less concerned about adding value in their investments, but more about contributing to the development of certain areas or industries where private-sector PEOs are less interested. Thus, a government-related PEO can devote less time to the follow-up of each investment, and this can have consequences on the final return obtained by the fund. In fact, Manigart et al. (2002) find that government-supported funds operate with lower required returns.

Regarding the proportion of divestments carried out through IPOs and trade sales over the total volume divested, private-sector-related PEOs record a higher average volume. The origin of this result may be found, again, in the different objectives sought by each type of PEO. While private-sector PEOs seek investments with a high probability of obtaining substantial capital gains, government-related PEOs may be involved in less profitable investments, which are more difficult to divest at a later date. The different size and growth patterns of their investments imply exit strategies focused mainly on management and/or owners buy-back. Regarding the amount of funds pending allocation, the median for the whole sample amounts to € 3.7 million, with similar values for both private and government-backed PEOs. However, average values are much higher for private PEOs.

Table 3
Descriptive statistics of the numerical variables related to the private equity activity (in 1991 € thousands)

Variables		Mean	Median	Standard deviation	Minimum	Maximum
Investments	Whole sample	4970.9	920.7	12,613.4	0	127,653.2
	Private PEOs	5808.6	793.0	14,542.8	0	127,653.2
	Government PEOs	3344.6	1070.1	7334.1		71,433.2
		<i>p</i> -value:*	(0.001)			
Ratio investments per manager	Whole sample	2.98	2.50	2.58	0	18
	Private PEOs	2.38	1.67	2.43	0	15
	Government PEOs	4.15	3.57	2.46	0	18
		<i>p</i> -value:*	(0.000)			
Divestments by IPO and Trsl (%)	Whole sample	0.153	0	0.312	0	1
	Private PEOs	0.193	0	0.352	0	1
	Government PEOs	0.081	0	0.203	0	1
		<i>p</i> -value:*	(0.000)			
Available capital	Whole sample	15,225.7	3726.4	37,196.5	0	442,336
	Private PEOs	18,657.0	3854.9	44,259.8	0	442,336
	Government PEOs	8691.3	3675.1	15,032.4	0	82,241
		<i>p</i> -value:*	(0.000)			

*Significance based on a *t*-test unilateral of equality of means.

Table 4
Descriptive statistics of new funds raised, distinguishing by different types of PEOs (in 1991 € thousands)

Panel A: new funds raised for the whole sample

		Mean	Median	Standard deviation	Minimum	Maximum
All PEOs		6254.1	0	25,412.9	0	455,865.1
National private equity association	Ascription	7177.3	0	29,056.1	0	455,865.1
	Non-ascription	4256.4	0	14,570.5	0	131,601.3
		<i>p</i> -value:*	(0.033)			
Government's private equity programme (Law)	Enrolment	6636.4	0	23,130.9	0	276,221.3
	Non-enrolment	5918.5	0	27,283.6	0	455,865.1
		<i>p</i> -value:*	(0.350)			
Location of headquarters	Capital city	11,991.3	0	37,129.0	0	455,865.1
	Other	2128.0	0	8944.3	0	131,601.3
		<i>p</i> -value:*	(0.000)			
Managers' experience	0–10 years	5357.4	0	16,890.8	0	150,217.8
	>10 years	8091.1	0	37,195.2	0	455,865.1
		<i>p</i> -value:*	(0.136)			
Size of private equity organization	Large	23,129.0	0	52,703.1	0	455,865.1
	Small	2160.6	0	6815.8	0	88,183.0
		<i>p</i> -value:*	(0.000)			

Panel B: new funds raised by different types of PEOs

	Mean	Median	Standard deviation	Minimum	Maximum
Private PEOs	8008.8	0	30,439.5	0	455,865.1
Government PEOs	2847.4	0	9198.8	0	82,636.7
	<i>p</i> -value:*	(0.000)			

*Significance based on a *t*-test unilateral of equality of means.

All the *t*-statistics for a difference in the average of each variable between private and government-related PEOs, reject the equality of means hypothesis.⁵

In Panel A of Table 4 some descriptive statistics of the endogenous variable are shown for the whole sample. The average volume of funds raised is € 6.25 million annually. This figure shows an important amount of dispersion, 0 being the minimum value and € 455.9 million the maximum one. The cyclical nature of private equity markets explains that the median has null value, since operators usually activate this process every 2 or 3 years. In fact, the average number of years prior to raising a new fund is 2.88 for the whole sample. Although not reported here, the percentage of PEOs raising private equity funds by year varies between 47.73% in 1991 to 38.46% in 2003, with a maximum of 49.06% in 1993. Thus, a high proportion of observations equals zero, as commented on in Section 4.

Considering the different characteristics of PEOs analyzed, it can be observed that the PEOs who raise most funds are those who are members of the national private equity

⁵ These tests assume different variances between both groups, as a variance equality contrast was made previously, with the homoskedasticity hypothesis being rejected in all cases except for the variable 'Ratio investments per manager'.

Table 5
Correlations of variables

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Experience	1.00												
2. Investments	0.09*	1.00											
3. Ratio inv/manager	0.31**	−0.14**	1.00										
4. Div. IPO and Trsl (%)	0.13**	0.23**	−0.06	1.00									
5. PE association	0.23**	0.12**	0.08*	0.18**	1.00								
6. Law	−0.10*	−0.04	0.03	0.07 [†]	0.25**	1.00							
7. Capital city	−0.15	0.25	−0.28**	0.25**	0.06	0.00	1.00						
8. Large	0.11**	0.51**	−0.19**	0.24**	0.15**	0.08*	0.41**	1.00					
9. Available capital	0.08*	0.53**	−0.21**	0.22**	0.11**	−0.00	0.34**	0.54**	1.00				
10. GDP growth	0.18**	0.12**	−0.06	0.12**	0.08*	0.13**	0.06	0.13**	0.07 [†]	1.00			
11. Interest rate	−0.24**	−0.17**	0.09*	−0.08*	−0.14**	−0.26**	−0.10*	−0.24**	−0.12**	−0.48**	1.00		
12. Stock market return	0.05	−0.11**	−0.01	0.07 [†]	−0.01	−0.13**	−0.03	−0.09*	−0.04	0.19**	0.01	1.00	
13. Capital gains tax rate	−0.23**	−0.18**	0.10*	−0.10*	−0.11**	−0.26**	−0.10**	−0.22**	−0.11**	−0.52**	0.90**	0.04	1.00

[†] $p < .1$, * $p < .05$, ** $p < .01$.

association, are geographically located in Madrid and are classified as large PEOs. The difference in the volume of new funds raised is not significant, however, for those PEOs who have signed up to the Law of private equity and those who have not or for those who have more or less than 10 years of experience. Nevertheless, this situation could be explained by the growth of the private equity market in Spain in the period analyzed, with a sizeable number of new entrants being registered in most years. All the *t*-statistics for a difference in averages between each of the subgroups, given a different variance for each type within each subgroup,⁶ reject the equality of means hypothesis except for the variables ‘Law’ and ‘Experience’.

Panel B of the same table reports the average volume of fundraising for both the private and the government-related PEOs. There is a considerable distance between the averages of both groups. The average amount of fundraising and the standard deviation is very much higher for private than for government-related organizations.

Table 5 presents the correlations among the independent variables. These correlations provide some insight into the univariate relations between the variables. Although the correlations indicate potential collinearity problems across some variables, such as interest rates and capital gains tax variables, the variance inflation factors were computed for each variable, ranging from 1.04 for Capital city to 5.82 for Tax. Therefore, multicollinearity is not an issue as they are all below the concern level of 10 (Von Eye and Schuster, 1998; Kennedy, 2003). Moreover, for all variables we checked robustness by including/excluding those variables jointly and separately and results do not change. Thus results shown in this paper are robust to different specifications.⁷ Therefore, all explanatory variables are included in the empirical analysis.

6. Results

Results for the whole sample are presented in Table 6. This table includes the results for the selection equation (probit) and the main equation (generalized least squares). The regressors included in the selection equation are the ones considered in Section 4. These same regressors—except four variables—and the terms *s*1 and *s*2, are included in the main equation. First, the variables GDP growth, interest rates and stock market returns are excluded from the analysis, as in Gompers and Lerner (1998). Second, the amount of capital still available for investment is excluded because it is considered that it has an impact on the probability of raising funds but not on the amount raised.

Regarding the set of variables related to the private equity activity, no evidence of the influence of experience on fundraising is found. This result was anticipated in the descriptive analysis, as the difference in the volume of funds raised was not significant between PEOs with over and under 10 years of experience. However, evidence is found of the effect of the volume of lagged investments on the volume of funds raised. This shows that the investment abilities of PEOs are positively perceived by investors. With regard to the ratio of portfolio companies to each investment manager in the previous period, evidence of a

⁶ Previously a variance equality contrast was made between each type and for each subgroup, with the homoskedasticity hypothesis being rejected in all cases except for the ‘Law’ group.

⁷ Other models not provided here are available from the authors upon request.

Table 6

Regression results for all private equity organizations

Independent variables	Dependent variable	
	First stage—probability of raising funds	Second stage—volume of funds raised
Experience _{it-1}	7.47E-4	411.01
Investments _{it-1}	6.73E-6	1.250***
Ratio investments/manager _{it-1}	-0.067**	-4348.79***
Divestments by IPO and Trsl% _{it-1}	0.296	21,963.68***
Private equity association _{it}	0.482***	24,254.31***
Law _{it}	-0.076	-4853.13
Capital city _{it}	-0.246	-4358.46
Large _{it}	0.865***	33,749.61***
Available capital _{it-1}	-1.88E-5***	—
GDP growth _{t-1}	2.869	—
Interest rate _{t-1}	4.153	—
Stock market return _{t-1}	-0.187	—
Capital gains tax rate _t	-0.884	-3765.58
s1 _i		2232.21***
s2 _{it}		29,717.88***
Constant	-0.631*	-79,787.77***
Log-likelihood	-378.56	
LR-test (<i>p</i> -value) ^a	0.000	
R ²		0.402

****p*<.01, ***p*<.05, **p*<.1.

Two-step Heckman methodology. The first stage is a random effects probit model and the second stage is a linear random effects model. The terms *s1_i* and *s2_{it}* are introduced in the second stage in order to correct the sample selection bias problem.

^aLR-test comparing random effects probit model with pooled probit model.

negative and significant impact both on the probability of raising funds and on the volume of funds raised is found. This result is consistent with the evidence presented by [Osnabrugge and Robinson \(2001\)](#), who interpret the follow-up of a smaller number of investments as indicative of a greater diligence in carrying out functions, thus signaling a better reputation. Finally, and regarding the exit process, the percentage of the volume of divestments through IPOs and trade sales over the total volume divested exerts a significant and positive impact on the volume of funds raised. Thus there is a strong positive signal about the PEOs associated with how diligently they carry out their investment and divestment activities. This shows that investors in developing markets believe that these indicators are signals of the future returns that PEOs will obtain and therefore, of their reputation, and this allows PEOs to raise more funds.

Within the group of variables related to exogenous signals, evidence is found of the impact of membership of the ASCRI Association both on the probability of raising funds and on the volume of funds raised. The impact of the variable *Law*, however, is not significant, which could be explained by the large percentage of funds that come from foreign investors who, in most cases, prefer to adopt legal vehicles they feel more familiar with. Similarly, the location of the PEO's headquarters in the capital city does not provide a useful signal to investors. [Gorman and Sahlman \(1989\)](#) find that most venture capitalists

Table 7
Regression results for private-sector and government-related private equity organizations

Independent variables	Dependent variable			
	Private-sector PEOs		Government-related PEOs	
	First stage— probability of raising funds	Second stage— volume of funds raised	First stage— probability of raising funds	Second stage— volume of funds raised
Experience _{it-1}	-0.024	-272.40	0.011	-247.59
Investments _{it-1}	1.03E-6	0.966***	7.98E-5**	1.066***
Ratio investments/manager _{it-1}	-0.049	-4853.22**	-0.094*	-154.41
Divestments by IPO and Trsl% _{it-1}	0.331	27,151.53**	0.265	6133.68
Private equity association _{it}	0.484**	23,690.97**	0.359	4753.06
Law _{it}	-0.084	-6257.47	0.052	-615.63
Capital city _{it}	-0.175	-7272.05	-0.155	4191.05
Large _{it}	0.810***	50,238.29***	0.815	8351***
Available capital _{it-1}	-1.69E-5***	—	-2.07E-5	—
GDP growth _{t-1}	1.917	—	3.891	—
Interest rate _{t-1}	0.868	—	8.954	—
Stock market return _{t-1}	-0.096	—	-0.186	—
Capital gains tax rate _t	-0.993	-68,878.89	-0.931	6447.90
s1 _i		5129.60***		93.19
s2 _{it}		48,391.39***		3413.42**
Constant	-0.342	-96,960.48***	-1.029	-14,585.35
Log-likelihood	-232.32		-138.46	
LR-test (<i>p</i> -value) ^a	0.002		0.147	
R ²		0.450		0.458

****p*<.01, ***p*<.05, **p*<.1.

Two-step Heckman methodology. The first stage is a random effects probit model and the second stage is a linear random effects model. The terms *s1_i* and *s2_{it}* are introduced in the second stage in order to correct the sample selection bias problem.

^aLR-test comparing random effects probit model with pooled probit model.

spend more than half their time monitoring their portfolio companies. PEOs usually locate their headquarters near the firms they invest in so as to provide greater assistance. Thus, the decision to locate PEO's headquarters could just be based on reasons of monitoring rather than on signaling reputation. These results show that among all of the external signals, investors only consider membership of the ASCRI Association as a signal of PEO's reputation.

Regarding the control variables, the coefficient of the capital under management is positive and significant. Thus, as in Gompers and Lerner (1998), larger PEOs raise larger amounts of funds. The amount of money available for investment shows a negative impact on the probability of raising funds, as expected. It should be noted that PEOs could not justify new fundraising when they have plenty of money pending allocation. However, no evidence is found of the impact of GDP growth (as in Jeng and Wells, 2000; Martí and Balboa, 2004), interest rates, stock market returns or capital gains tax rate on fundraising. Finally, it should be remarked that both *s1* and *s2* terms are significant, so the Heckman methodology is more appropriate than the Tobit technique.

Table 7 presents the results for private and government-related PEOs. In the case of the former the panel considered incorporates 79 PEOs (with 497 panel data observations), and in the latter it includes 26 organizations (with 256 panel data observations).⁸

The results show that while the number of significant variables is high in the group of private-sector PEOs, this is scarce in government-related PEOs. Regarding the variables related to the private equity activity, results show that for both types of PEOs the volume of lagged investment has a positive and significant impact on the volume of funds raised, thus verifying the importance of investments made as a signal of ability to manage a greater volume of resources. The ratio number of portfolio companies per investment manager has a negative impact on the volume of funds raised for private PEOs and on the probability of raising funds for government-related PEOs. Specifically, it is found that investors are penalizing those PEOs holding an excessive number of portfolio companies per investment manager. The signal related to the percentage of divestments made through IPOs and trade sales is only confirmed for private-sector PEOs, so investors allocate more money to PEOs who divest through these ways.

Regarding the external signs, different signaling patterns can be found between private and government-related PEOs. ASCRI membership has a positive impact on the amount of funds raised, but only for private-sector PEOs. The amount of funds under management exerts a positive impact on the volume of funds raised for both types of PEOs, but also on the probability of raising funds in the case of private PEOs. The lagged volume of capital available for investment has a negative impact on the probability of raising funds but only for private-sector PEOs. Thus, results show that government-related PEOs, which get their funds from government organizations, may have less pressure on signaling reputation to the market than private ones. Finally, the significance of both *s1* and *s2* confirms the superiority of the Heckman estimation.

The models analyzed rely on past investments and divestments, among other variables, as a signal of reputation. These variables, however, are not relevant for new entrants in the industry, since they are not able to show a record of previous investment/divestment activity. **Table 1** shows that nearly half of the existing private-sector-related PEOs in 2003 were created from the year 2000 onwards, thus, introducing some doubts about the real importance of some variables related to the investment and divestment activity.

In order to check how the different signals would change when newcomers are excluded, we ran the regressions excluding data from new investors entering the field from 2000 onwards. This also allows us to test *Spence's (1973)* observation that signals must be credible. Regarding private-sector PEOs, an increase in the coefficient of the variables related to the investment and divestment activity is expected. Results show that these coefficients rise to 1.63 and 29,767.95, respectively. The coefficients of the other significant variables in **Table 7** are either more positive or negative, thus, providing sound evidence of a signaling activity when second and subsequent fundraising efforts are developed by existing PEOs. Regarding the government-backed PEOs, their number remains steady, so no significant changes in their coefficients are expected.

⁸ The sum of both does not come to 101 because during the period analyzed 3 government-backed PEOs became private ones and 1 PEO made the reverse change.

7. Discussion

The agency relationship between investors and private equity managers determines the need for the latter to establish a high reputation. This paper hopes to contribute to literature by considering how investors view several aspects of the activity carried out by private equity managers and some external variables as signals of PEOs reputation in markets characterized by high levels of information asymmetry. The results found in this paper show that some variables related to the private equity activity are important in this context. The amount invested has a positive impact on the volume of funds that will be raised by managers the following year. However, as the ratio of portfolio companies per investment manager increases, both the probability of raising funds and the amount of funds raised decrease because management attention to each investment may diminish and this could reduce future returns obtained by managers. There is also a positive impact of the percentage of divestments through initial public offerings and trade sales on the volume of funds raised, which is consistent with the fact that these mechanisms are the most profitable ways of exiting.

A variable related to external signs is also considered by investors. The PEO's ascription to the national venture capital and private equity association has a positive impact on both the probability of raising funds and on fundraising. In this way, investors consider that this signal sent by PEOs is credible, so they would commit more funds to those PEOs who have invested in this signal. Regarding the control variables considered, the size of the private equity organization has a positive impact on fundraising, while the lagged volume of funds awaiting investment shows a negative impact on the analyzed variable.

This paper also studies the different signaling patterns that could be used by private and government-related organizations. The volume of investments and the ratio number of investments per investment manager are signals used by both types of managers. However, operators funded by private sources also signal their reputation through the other fundamental task performed in the private equity cycle, such as the percentage of divestments made through IPOs and trade sales, and through membership of the national venture capital and private equity association. The volume of funds still available for investment is a significant control variable that explains fundraising in the private-sector group.

The fact that government-related operators scarcely send any signal was expected. The operators for whom the majority of resources is provided by private investors are more obliged to account for their performance to them. As a result, they have to be more concerned about building and maintaining a good reputation, since on that will depend, to a great extent, the volume of funds raised in the future. This could be the reason why the number of significant signals is greater for this group. In the case of government-related operators greater importance may be given to economic and social aims, as they were fundamentally set up to contribute to the development of certain geographic areas or sectors of specific interest.

There are meaningful implications for other developing countries from the evidence observed in the Spanish case, such as the Western European ones. The United Kingdom (UK) would be an exception, since, among these countries, it ranks first both in terms of volume of funds raised and invested, and is considered a consolidated market. Although European markets show some differences, regarding, for example, the venture capital intensity (Bottazzi and Da Rin, 2002) or the investment strategies of venture capitalists (Schertler, 2001), most of

them show similar trends, such as a decreasing importance of banks as capital providers (Schertler, 2001; Bottazzi and Da Rin, 2002), a steady growth in the size of funds, a concentration of investments on later stages or a predominance of trade sales as the mechanism used to divest (Christofidis and Debande, 2001). Moreover, and without considering the UK, the European venture capital industries have experienced similar patterns in terms of growth, as they all emerged in the 1980s, and most of them have experienced rapid growth in the second half of the 1990s. European venture capital still lacks human resources (Bottazzi and Da Rin, 2002), and PEOs in these countries are still very young and learning their trade. Most of them have not had the time to take many companies public and their track record is too short to help investors evaluate their abilities. Therefore, the signals proposed in this paper could be highly regarded by investors in these markets. As an example, a national sector-related association was created in the second half of the 1980s for almost all European countries, and all their members have to accept a code of conduct.

Especial mention should be made of the extension of the results obtained to the Eastern European countries, since their economies are nowadays experiencing a similar development to the one Spain experienced a decade or more ago. The empirical analysis in this paper covers this period for Spain, and coincided with the time soon after Spain joined the European Union. Some of these countries have recently joined the European Union, and their economies are becoming more market oriented, as occurred in the case of Spain. They are increasingly attracting the interest of investors, and their markets are expected to grow substantially in the coming years. Thus extending the results of this paper to the experience of these countries nowadays seems suitable and interesting. Moreover, the venture capital markets in these countries emerged later on in time, in the early 1990s, so the lack of PEO's track records is even higher and PEOs are forced to signal their reputation. Interesting future research would entail testing whether the development of their venture capital and private equity markets is based on similar variables as in the case of Spain.

The analysis conducted in this paper may also be a useful tool both for venture capital and private equity managers and investors. Regarding the former, they may now identify the characteristics that are considered by investors when choosing a fund in which to commit their money. This proves very important because the private equity activity forces managers periodically to raise fresh funds from investors to continue their activity. The study also offers a guide to managers that want to raise their first fund, since it has identified the effect that some variables not related to the private equity activity itself have on fundraising. It also has important implications for investors, as results found may be an important guide that allows them to distinguish between good and bad managers, helping them to allocate their money wisely.

Further research can address the effect that several well-known practices in this industry may have on the results obtained. First, venture capital and private equity operators usually specialize in certain industries or stages in order to exploit their technical expertise, to control portfolio risk and to better monitor their investments (Barry, 1994). The expected return could then be higher for funds which specialize in contrast to those that do not. Thus, specialization could also be a driver of reputation, which could also be of a greater importance in developing markets where track record on past funds is not available.

Second, they usually syndicate investments, which allows risk avoidance through risk sharing, brings together more expertise and support and, more importantly, may lead to a

superior selection of investments (Lerner, 1994). Brander et al. (2002) find that syndicated deals have higher returns. Hence, syndication may also have an impact on the reputation of managers.

Third, they typically invest in firms at different stages in their development. Staging the capital commitment allows them to preserve the right to abandon a project, to gather information, to monitor the progress of firms and to provide incentives to the entrepreneurial team (Sahlman, 1990). As this author argues, this is one of the most important mechanisms for controlling the investee firm. Then, it is interesting to study whether higher returns are obtained when the staging investment strategy is applied and its consequences on reputation.

Unfortunately, we were unable to check these three practices on this data set because under 5% of Spanish PEOs are specialized by industry, due to the small size and limited development of the domestic private equity market, and less than 10% of the amount invested is syndicated. Furthermore, the low percentage of investments devoted to early-stage companies also causes a marginal use of staging strategies (Martí, 2002).

Another further extension of the analysis performed here entails confirming whether the more reputed managers according to the characteristics found in this paper are those who really obtain higher returns. However, this could only be checked as data on returns become available, which is not yet the case for the Spanish market. Also, the results obtained in this paper could be confirmed by using data on other venture capital markets that are still in the process of maturing, as previously commented. Another interesting question relates to the factors that determine reputation in mature markets. According to the signaling theory, the importance of a signal is likely to be higher in markets with a high degree of uncertainty. Thus, are these factors really different from the ones identified here? More research on this topic is deserved.

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